

Advanced Embedded Systems

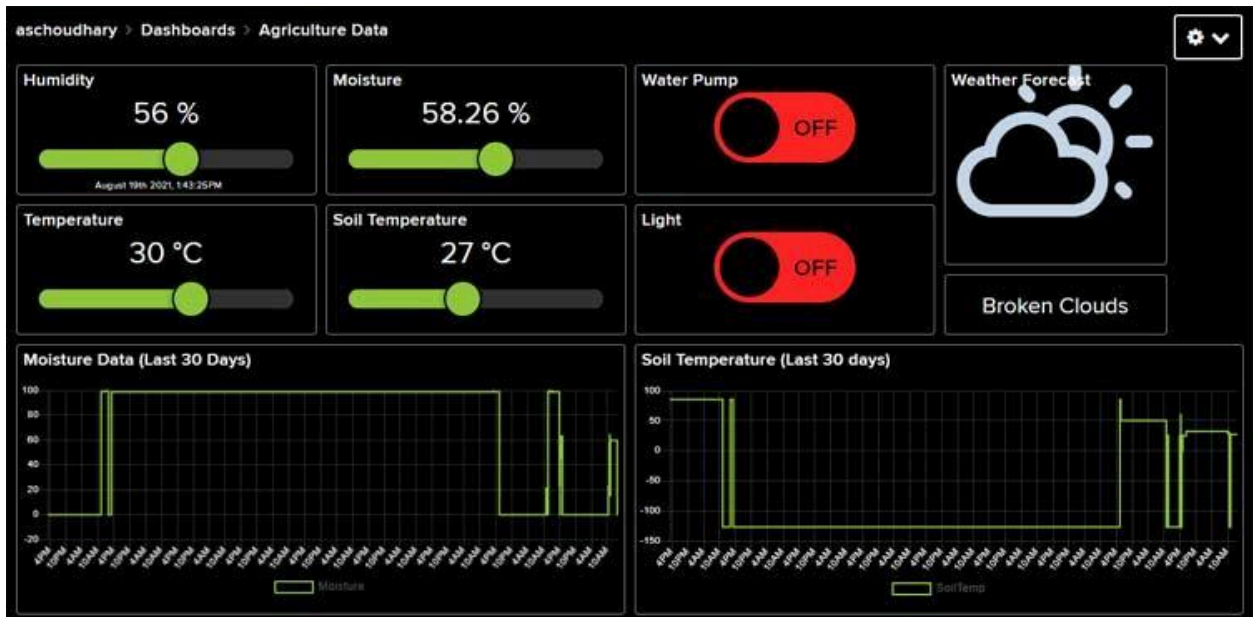
Project Ideas

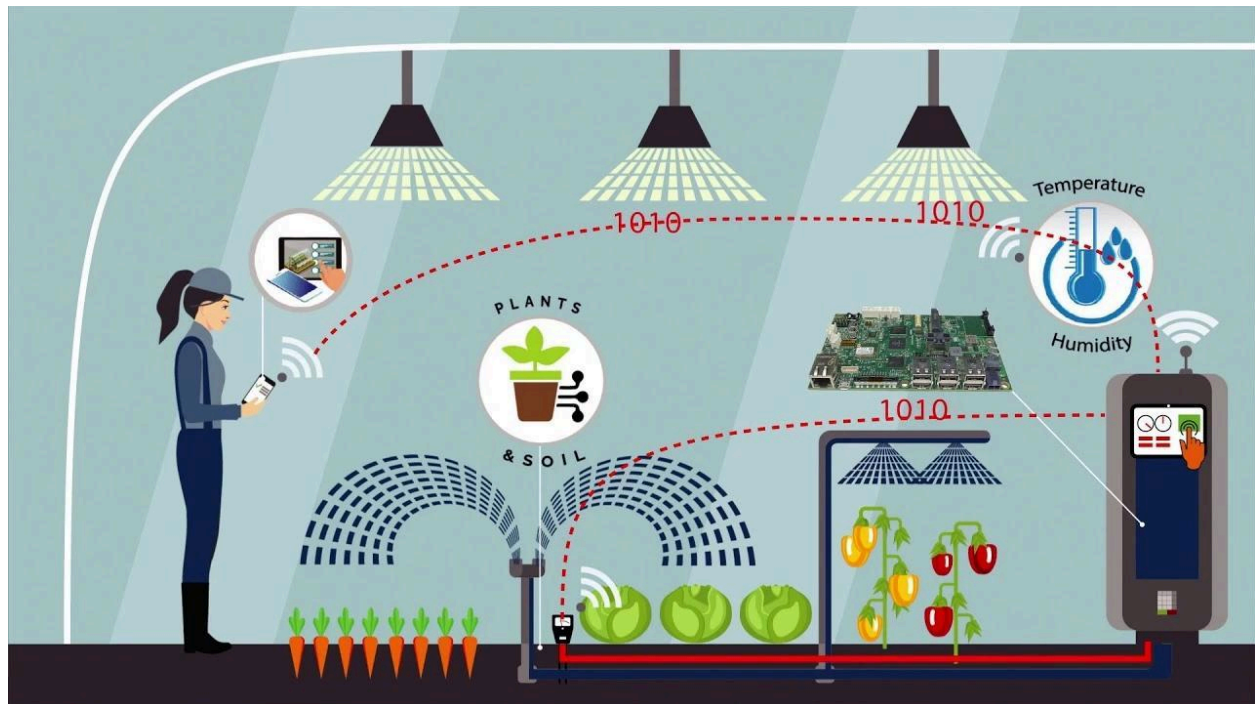
1. Smart Agriculture











Components

- ESP32
- Soil Moisture Sensor
- DHT11/DHT22
- Water Pump
- Relay Module
- LoRa/Wi-Fi

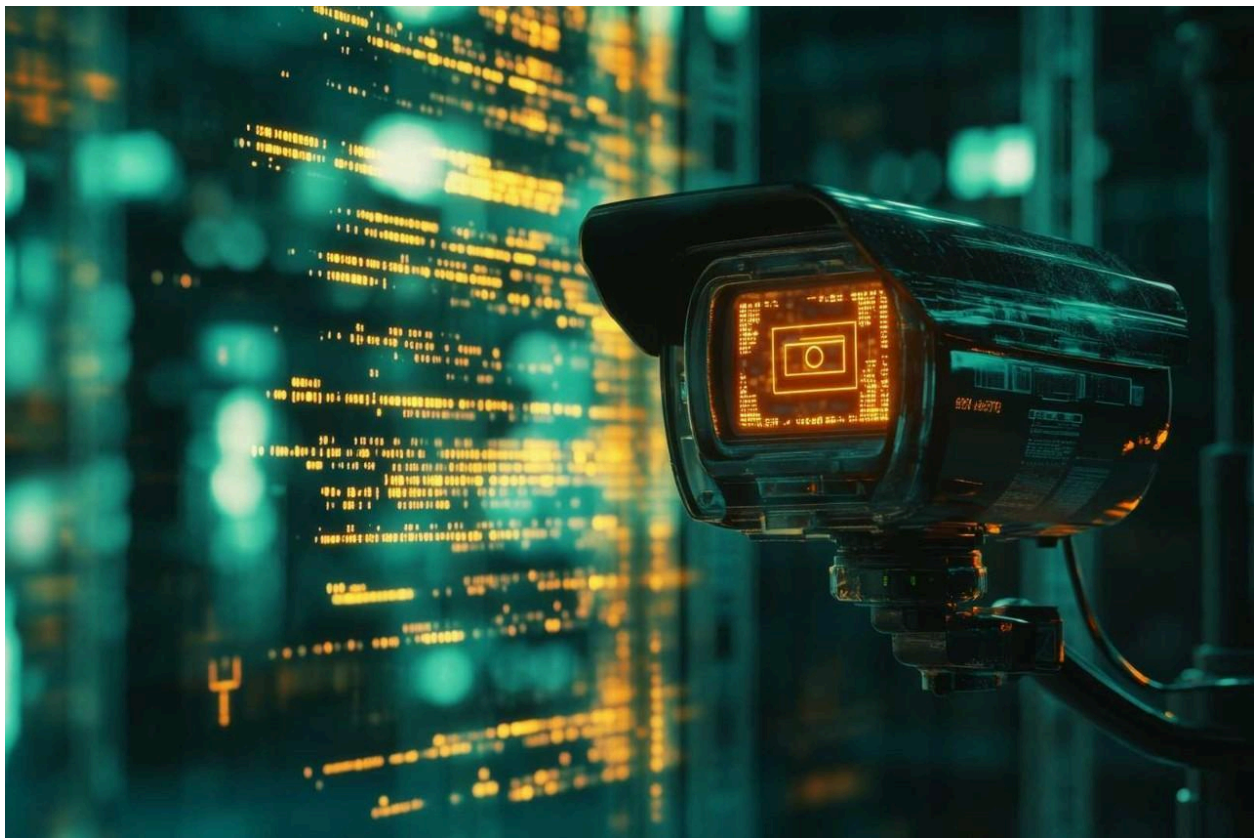
Features

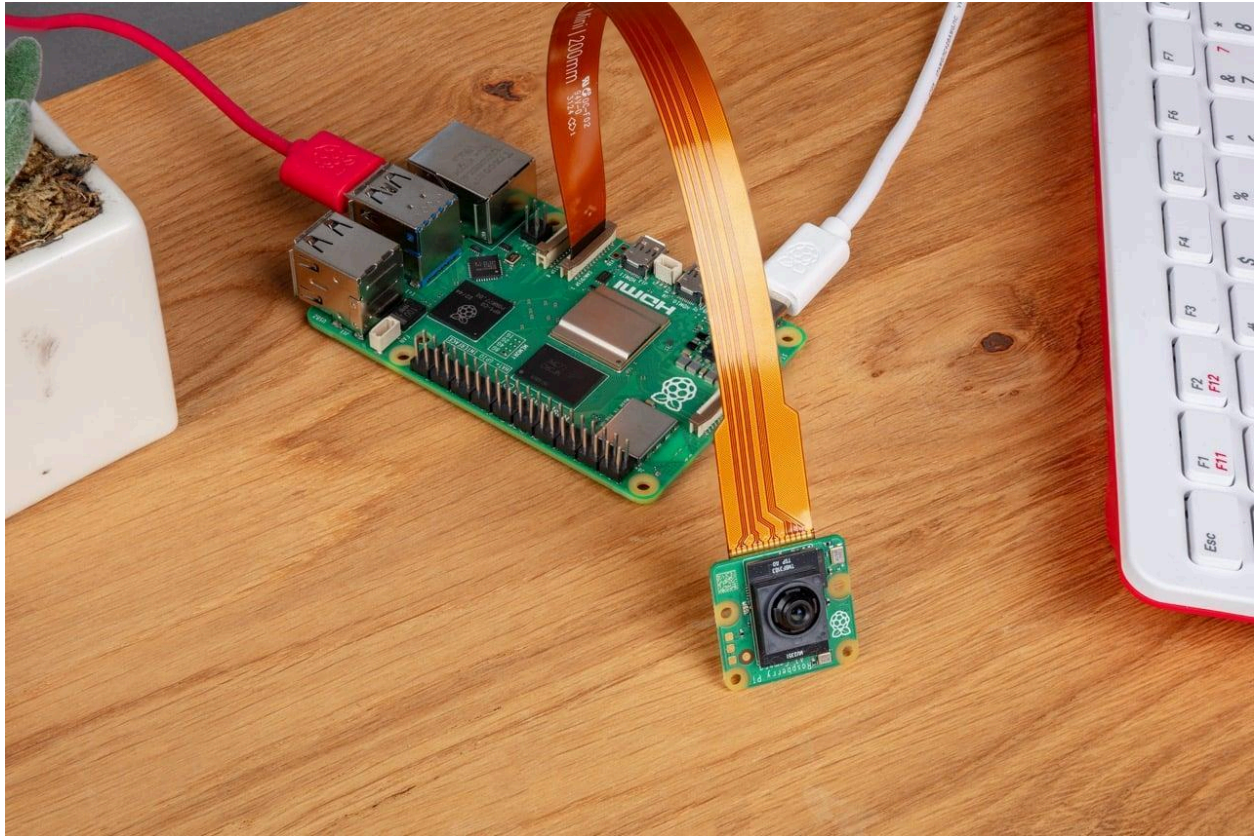
- Automatic irrigation
- Soil monitoring
- Mobile dashboard
- Weather-based watering

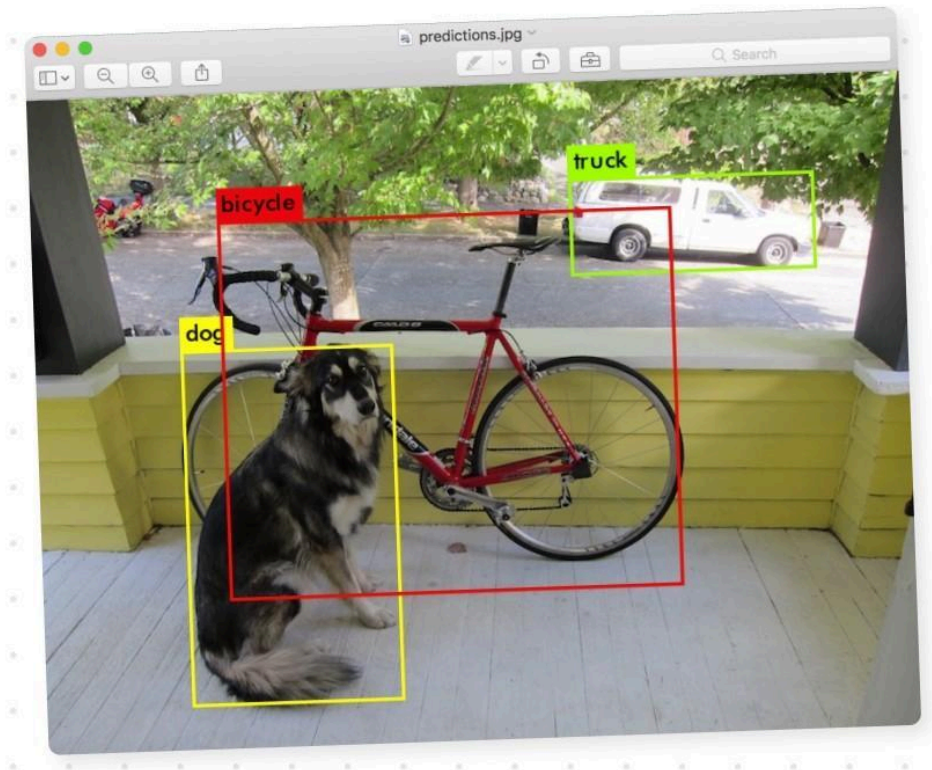
Skills Needed

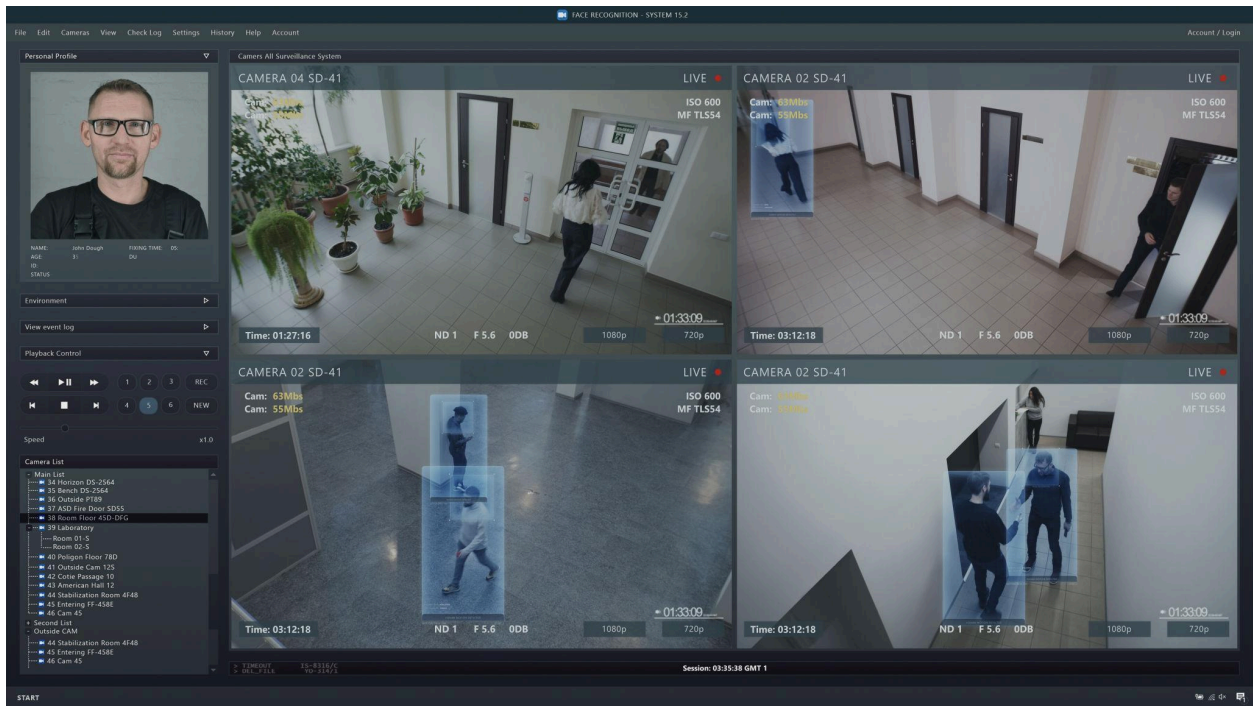
- IoT
 - Sensors
 - Embedded C/Python
 - Cloud integration
-

2. AI CCTV Camera















Components

- Raspberry Pi
- Camera Module
- OpenCV
- YOLO AI Model
- Motion Sensor

Features

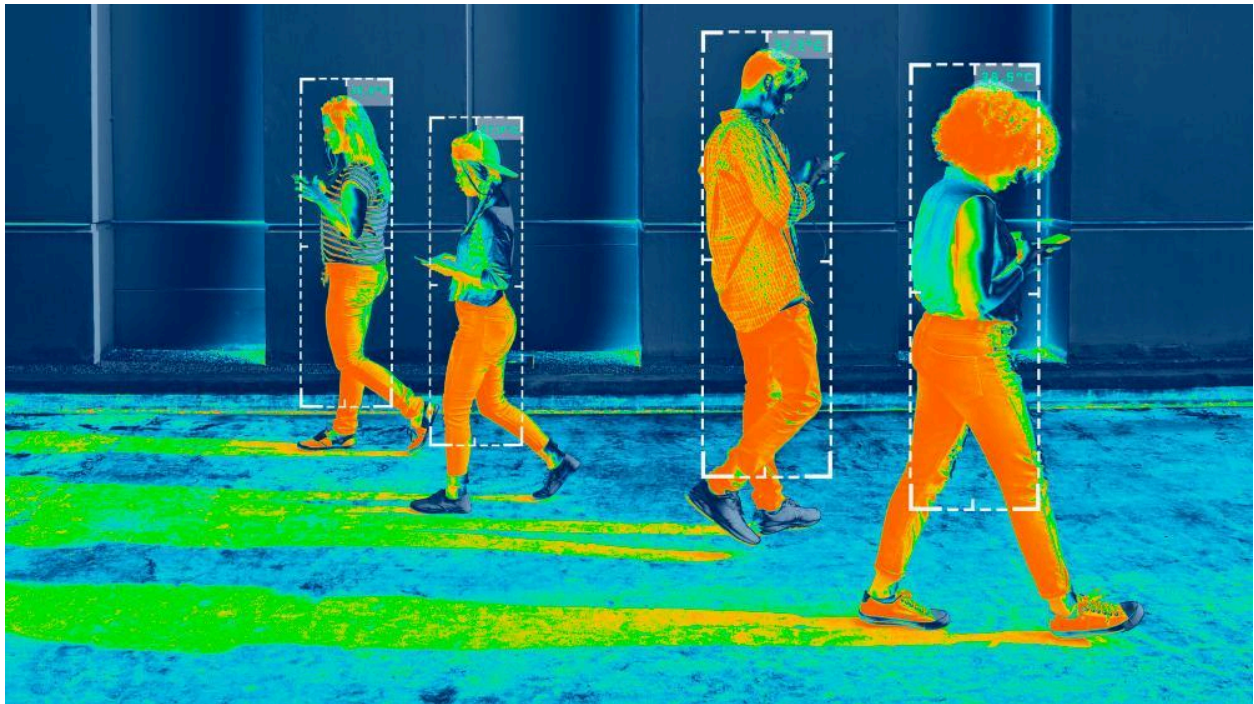
- Human detection
- Face recognition
- Intruder alerts
- Real-time streaming

Technologies

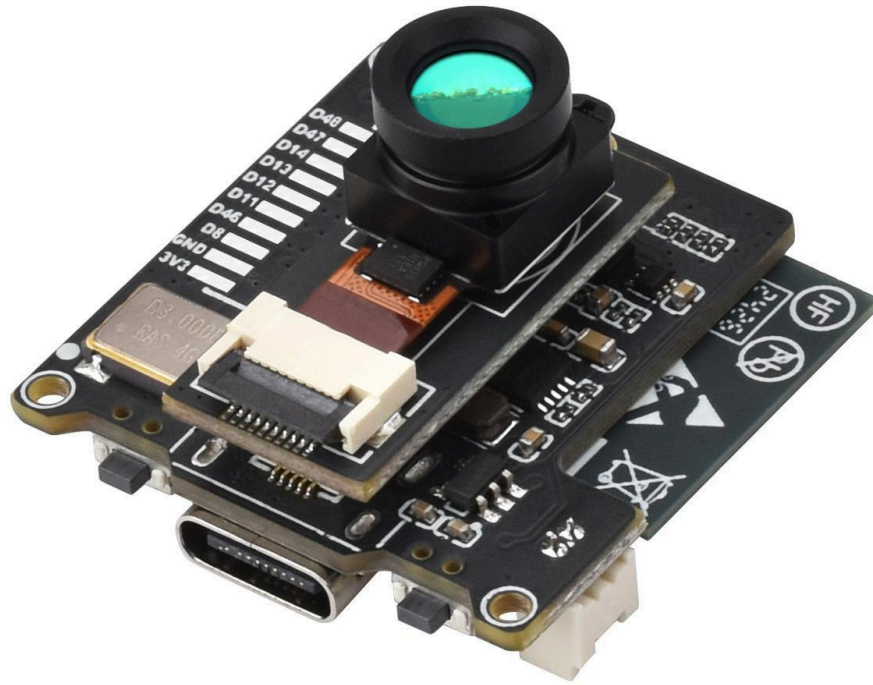
- Python

- OpenCV
 - Deep Learning
-

3. Thermal Human Detection













Components

- MLX90640
- ESP32 / Raspberry Pi
- OLED Display
- Wi-Fi Module

Features

- Detect humans in darkness
- Heat map visualization
- Intruder alerts

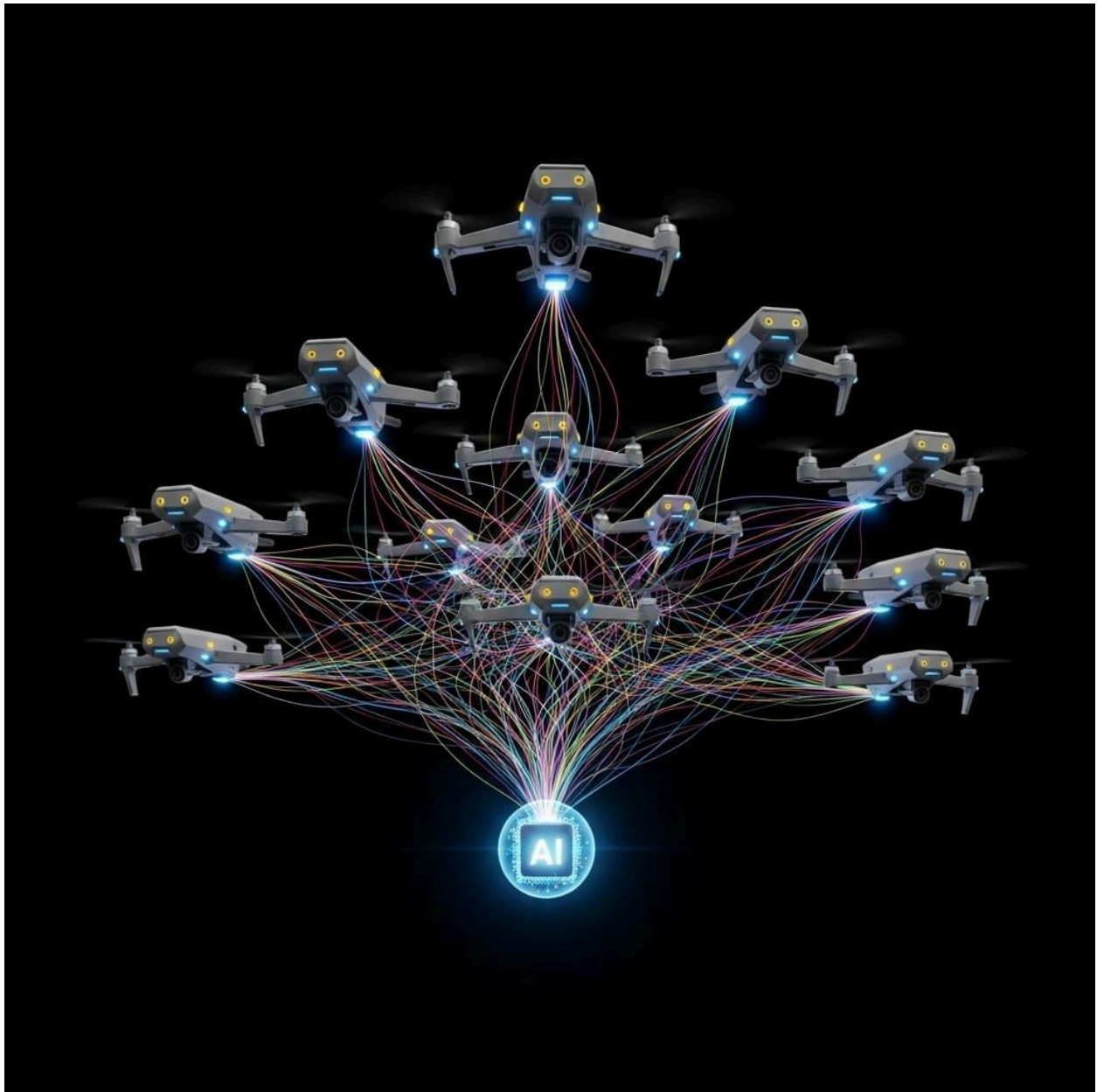
- Defense applications

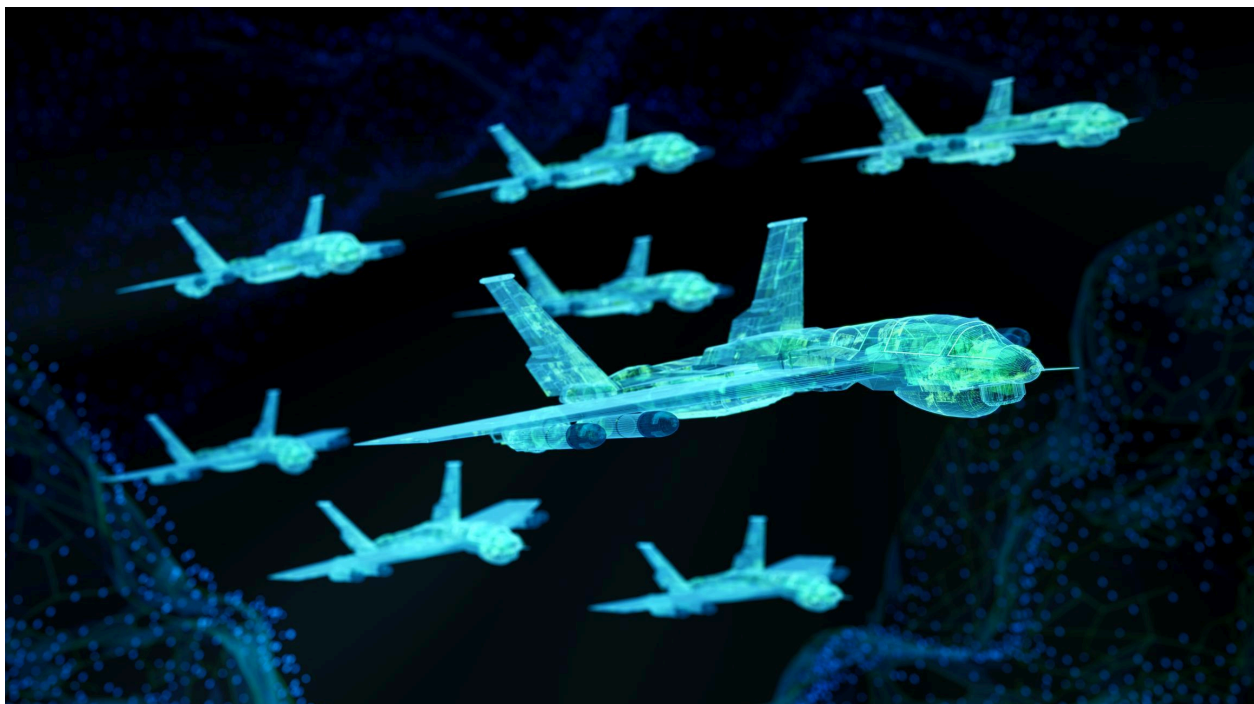
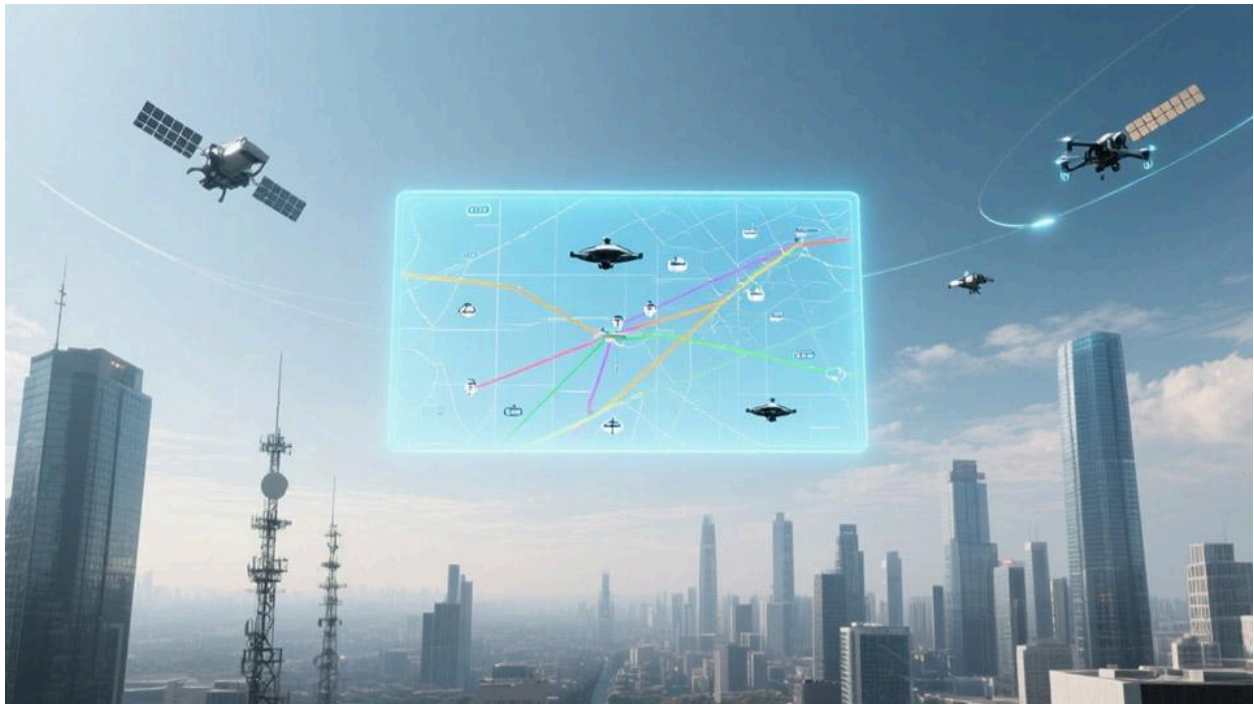
Future Scope

- Border surveillance
 - Rescue operations
 - Smart defense systems
-

4. Drone Swarm







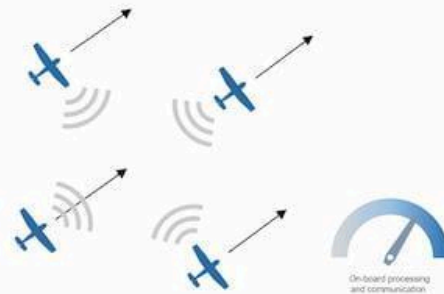


TYPES DRONE SWARMS AND HOW TO TELL THEM APART

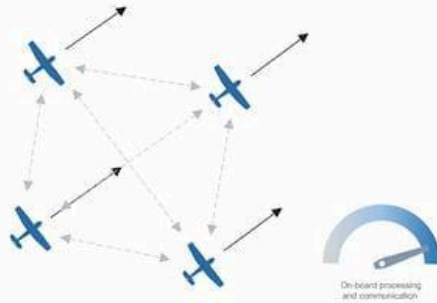
CENTRALIZED SWARM CONTROL



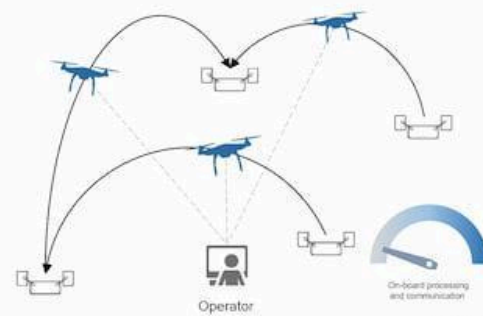
DECENTRALIZED SWARM CONTROL



DISTRIBUTED SWARM CONTROL



MULTIPLE DRONE OPERATION (NO SWARM)



Source: Drone Industry Insights, Advancement Challenges in UAV Swarm Formation Control: A Comprehensive Review

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DRONE
INDUSTRY INSIGHTS

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Components

- Flight Controller
- GPS Module
- LoRa/XBee
- Raspberry Pi
- IMU Sensors

Features

- Multiple drone coordination
- Formation flying
- Target tracking

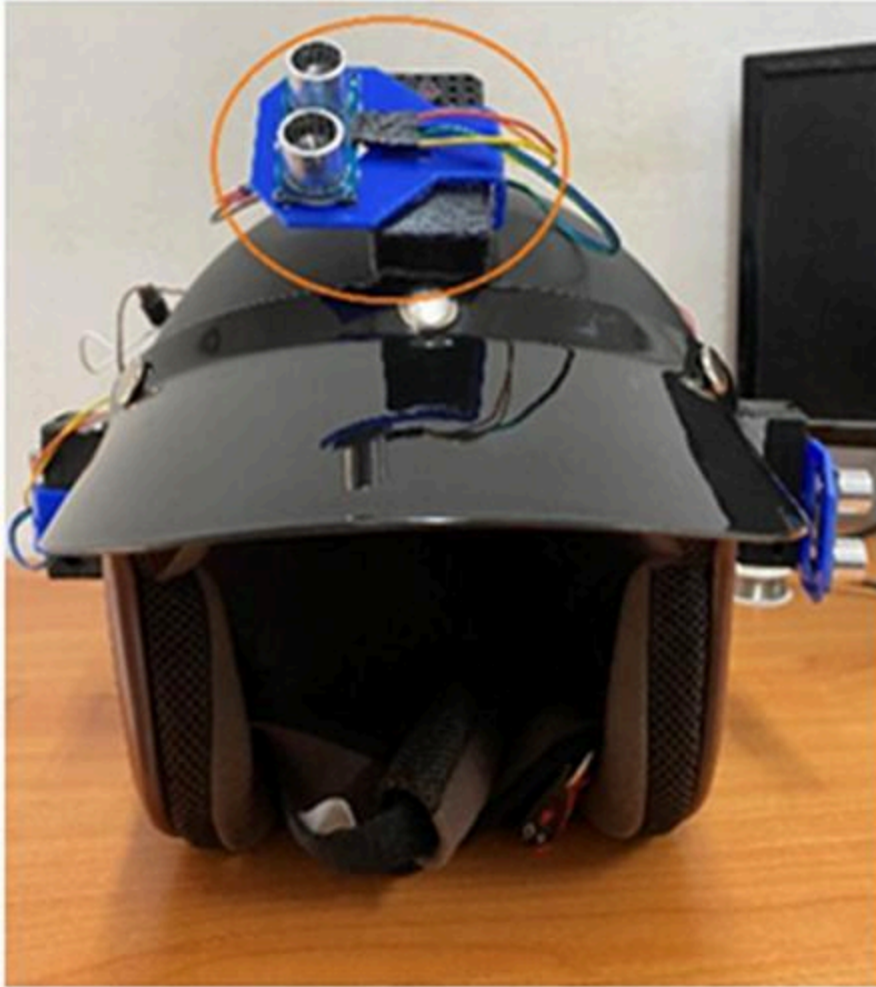
- Autonomous navigation

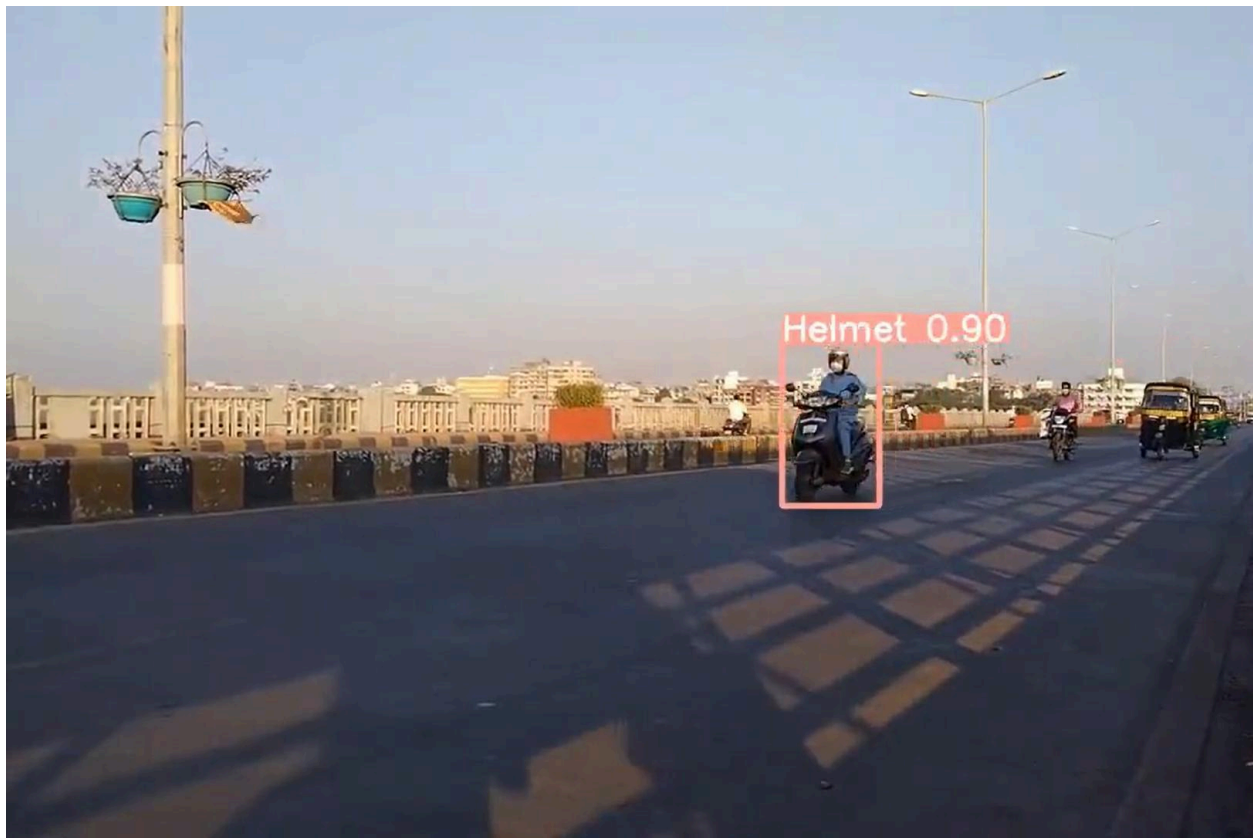
Skills

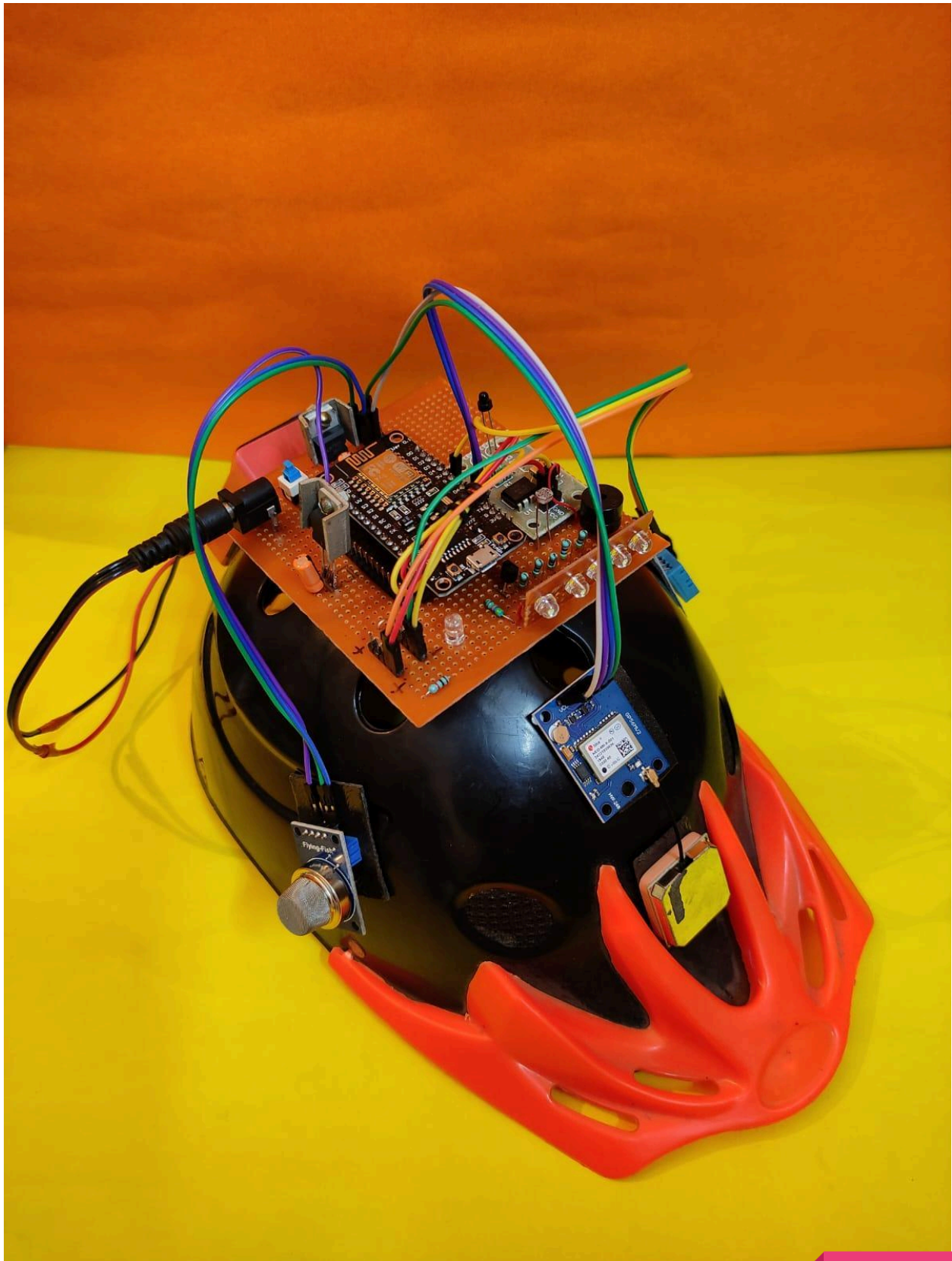
- Embedded Systems
 - Wireless Networking
 - AI
 - Robotics
-

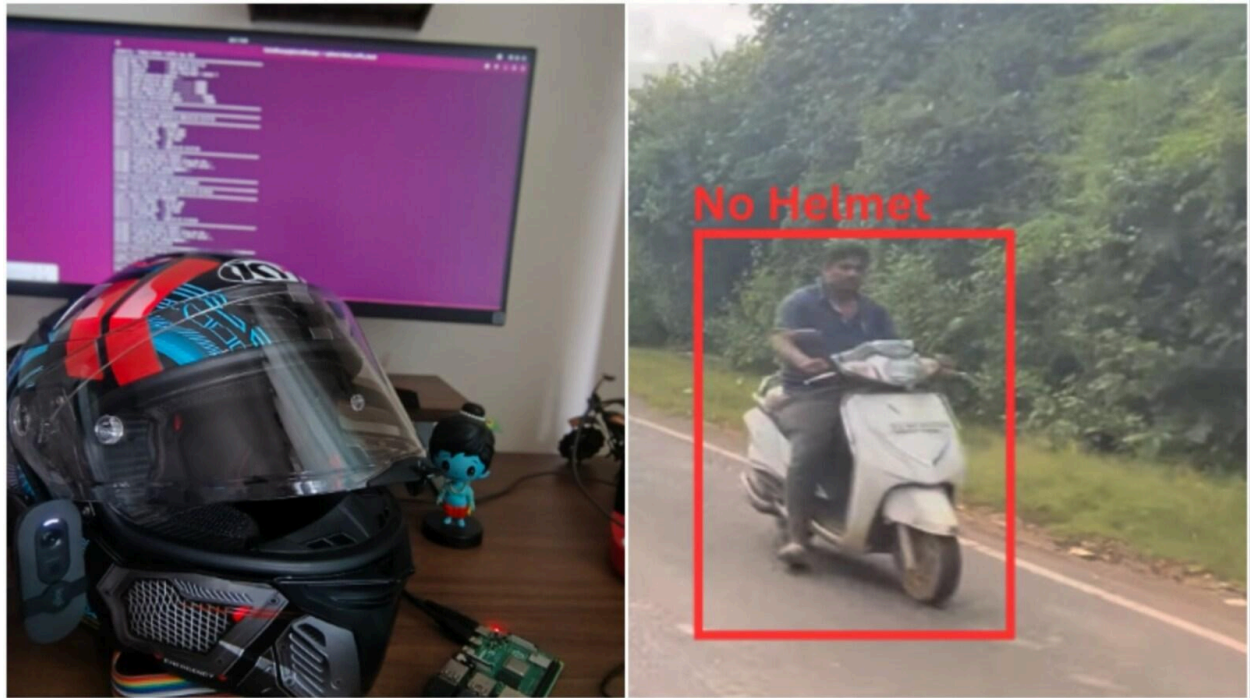
5. Smart Helmet











Block diagram

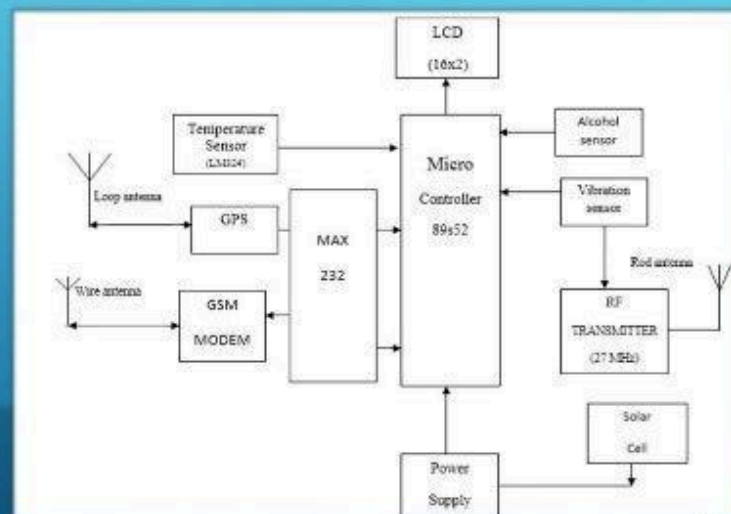


Fig.1 Block Diagram of Smart Helmet System.



Features

- Accident detection
- Alcohol sensing
- GPS tracking

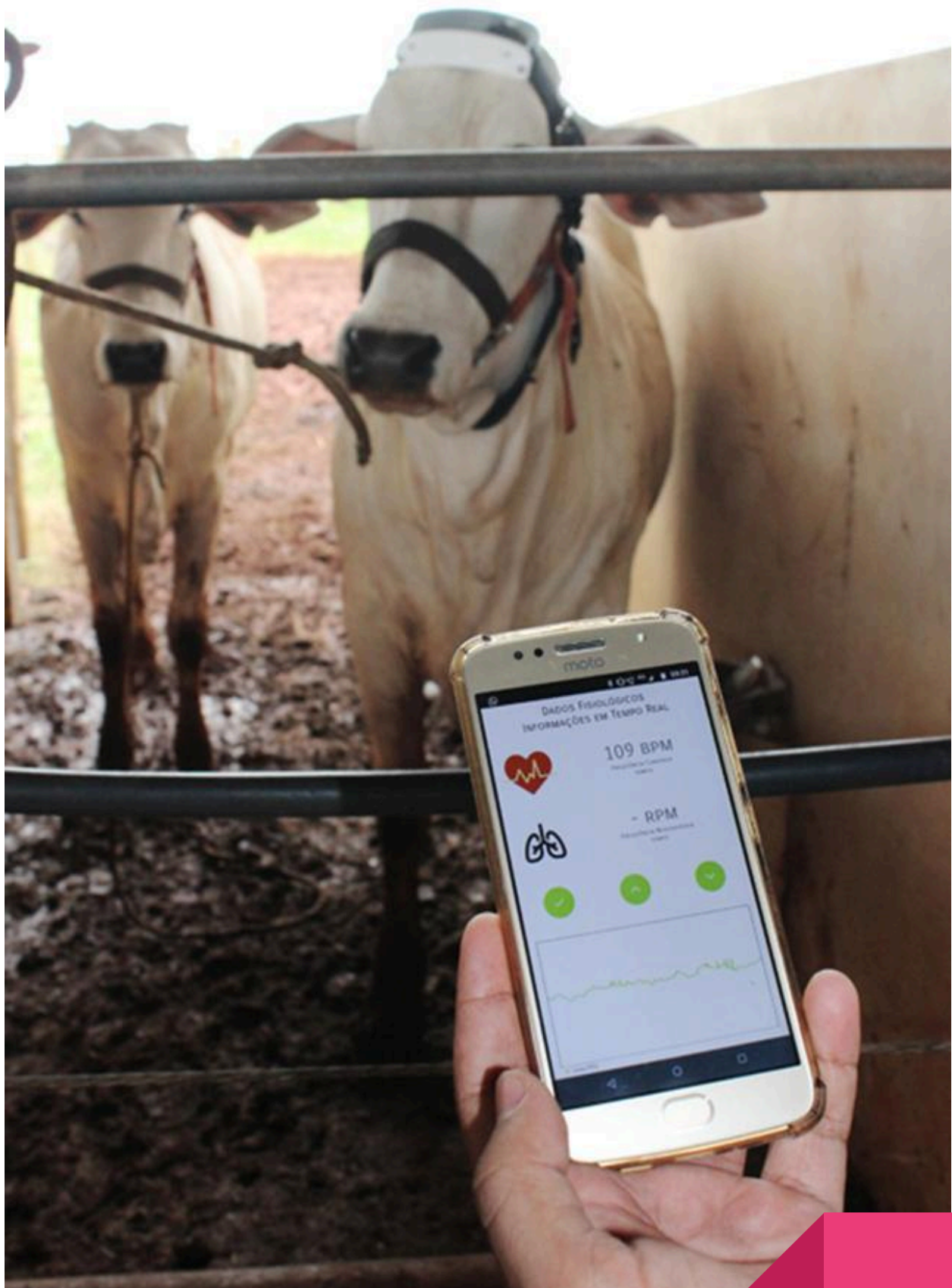
- Emergency SMS alerts

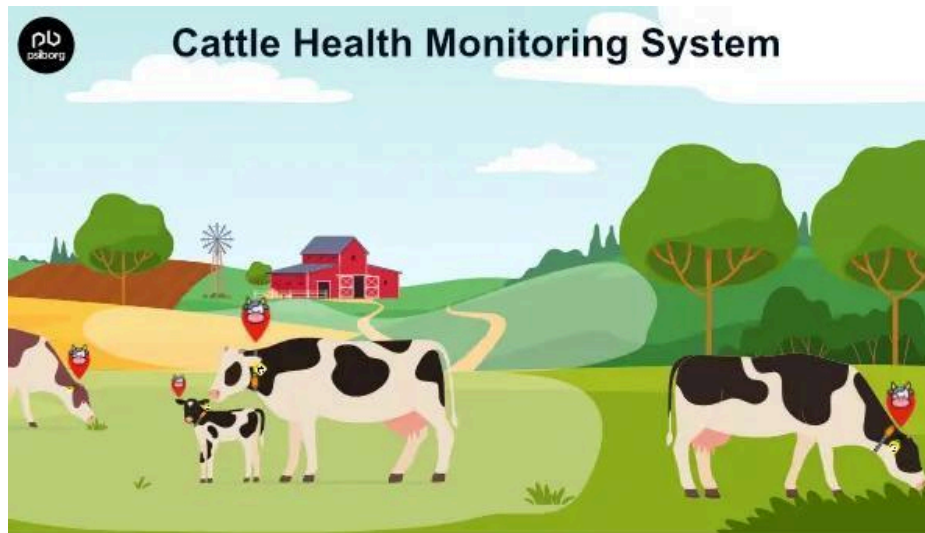
Components

- ESP32
 - MPU6050
 - GSM Module
 - Alcohol Sensor
-

6. Animal Health Monitoring

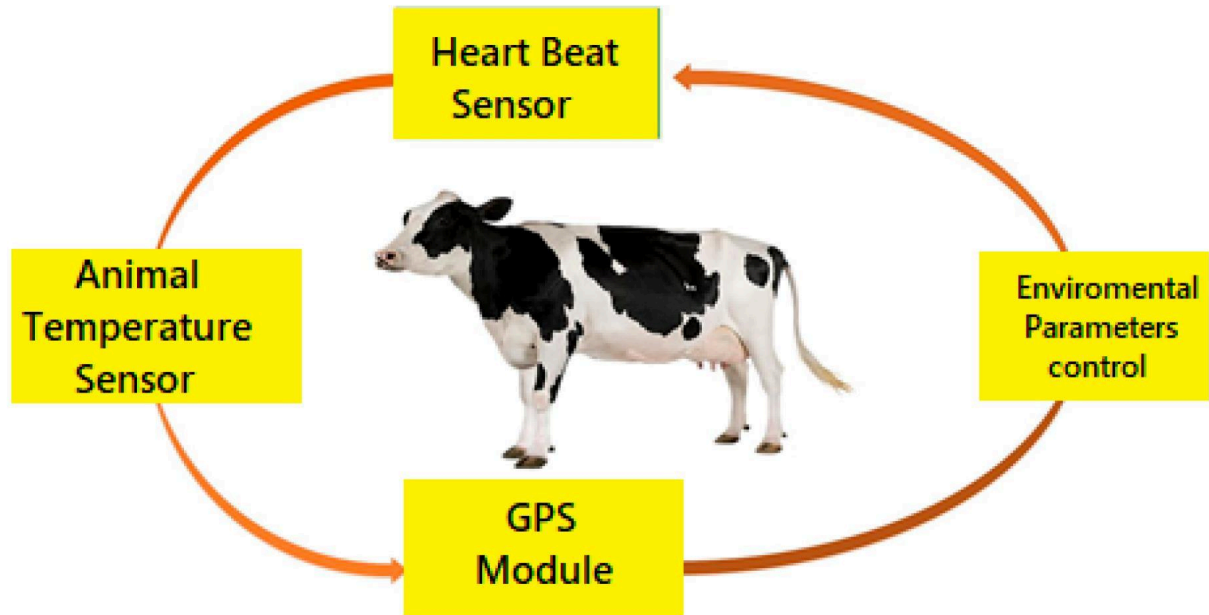


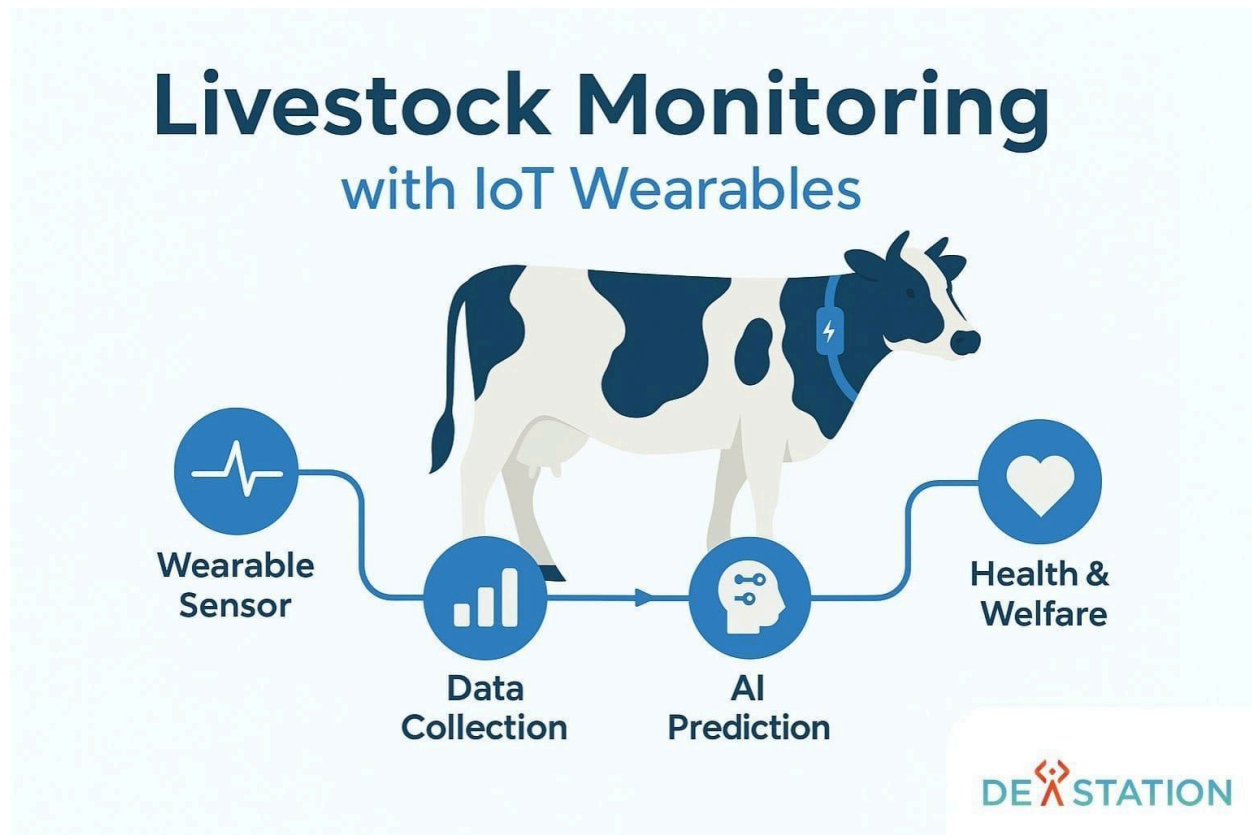












Features

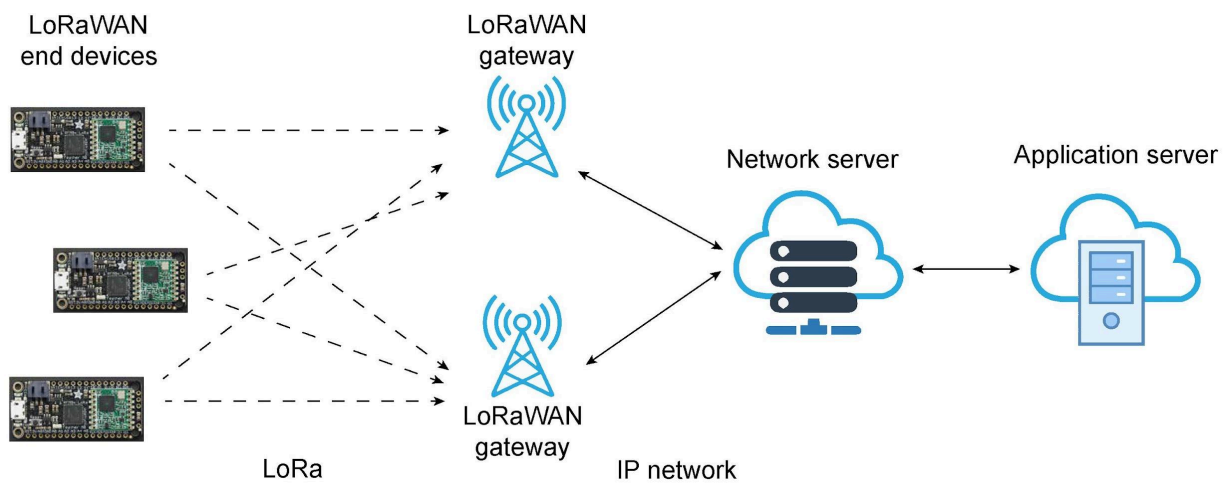
- Temperature monitoring
- Disease prediction
- GPS tracking
- Health alerts

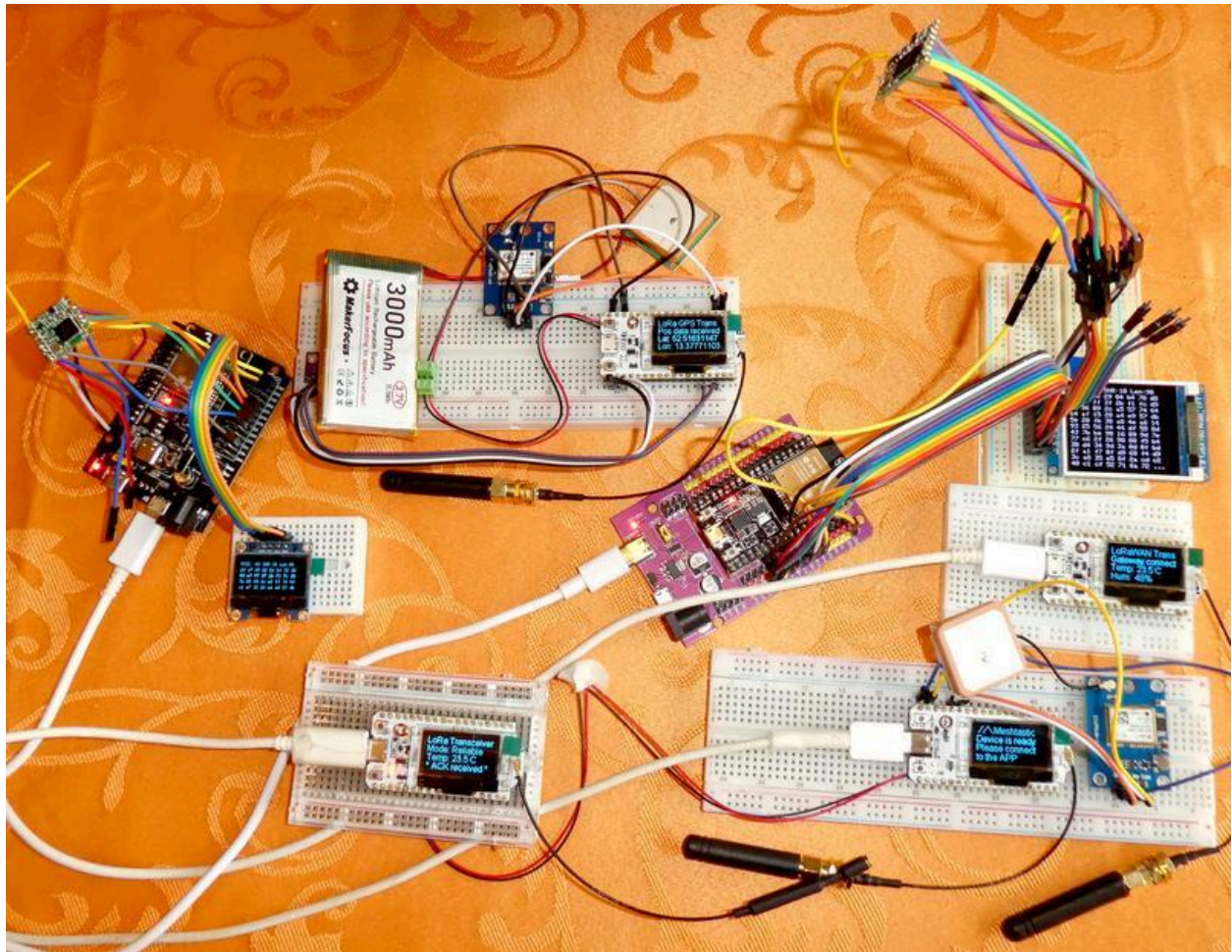
Components

- Temperature Sensor
- ESP32
- LoRa
- GPS Module

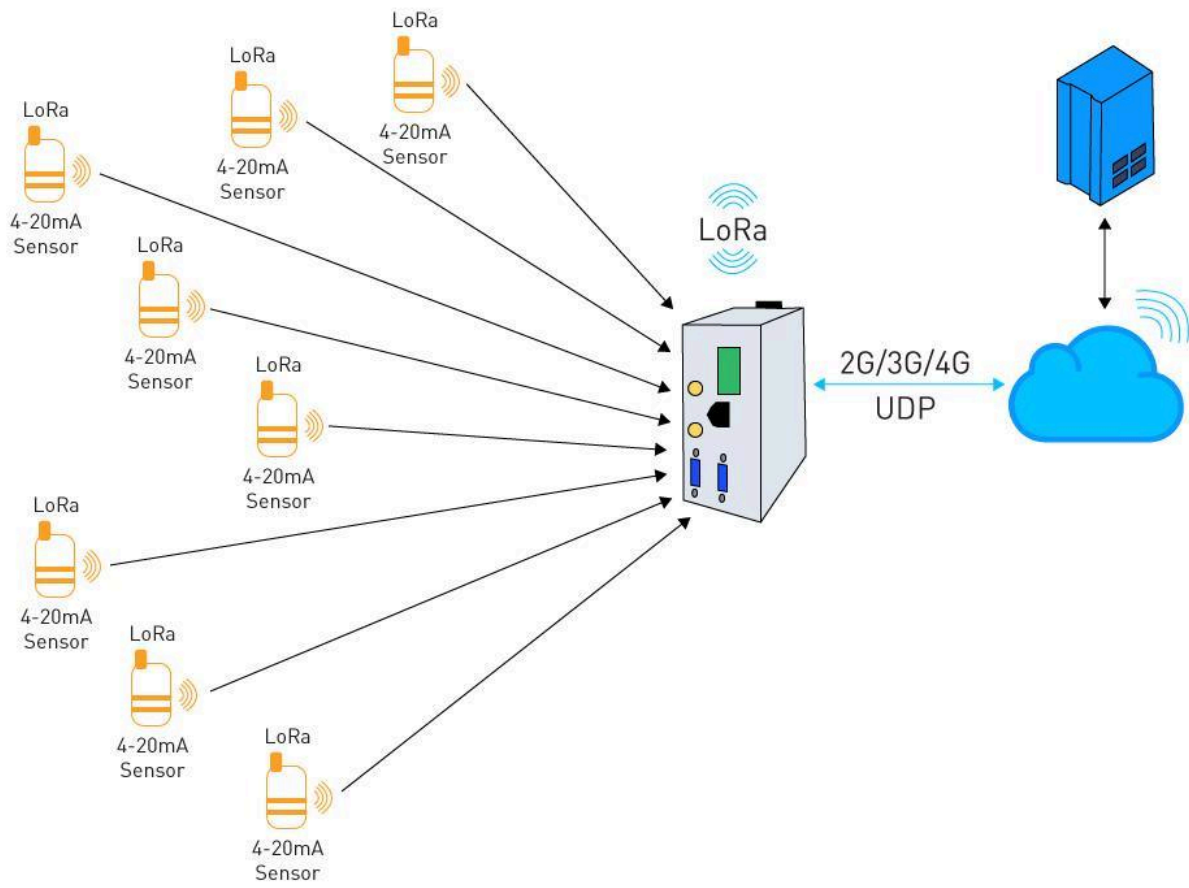
7. LoRa Communication System

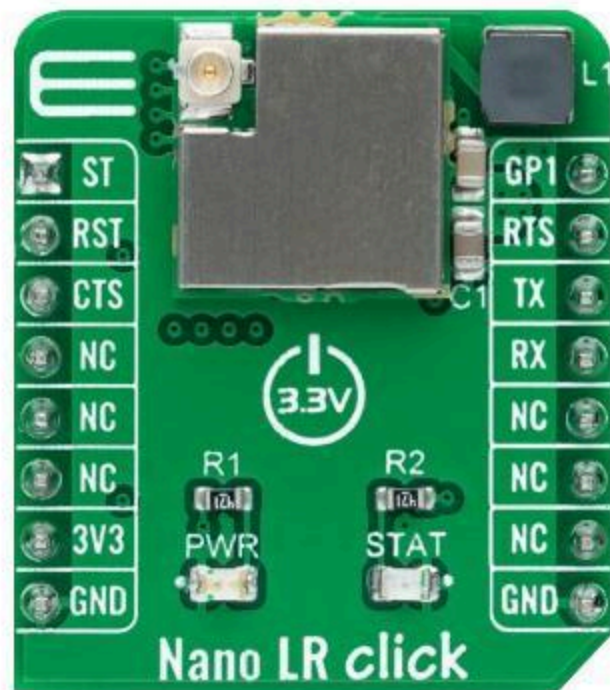














Features

- Long-range communication
- Low power usage
- Sensor networking
- Remote monitoring

Applications

- Smart villages

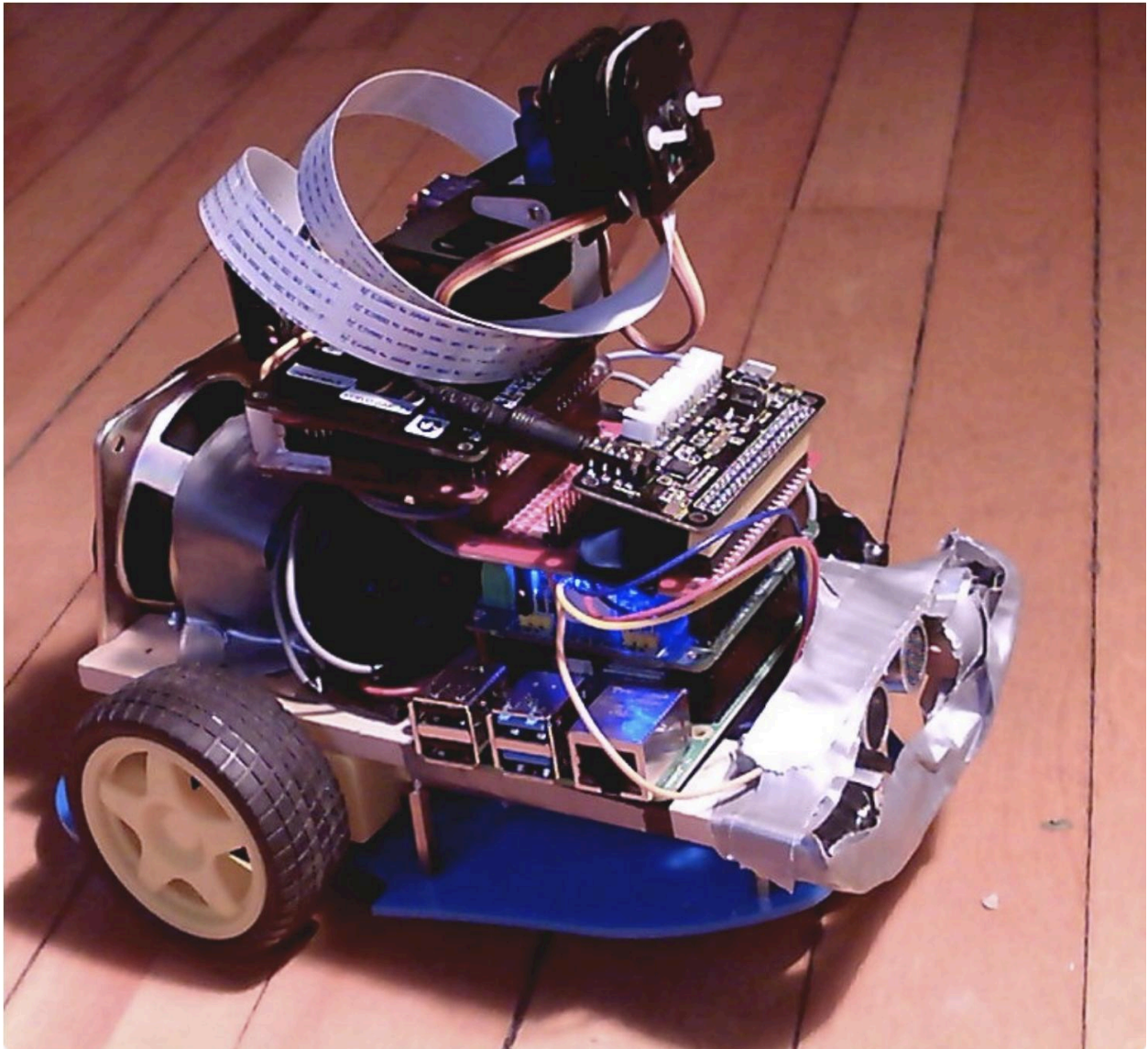
- Agriculture
 - Defense communication
-

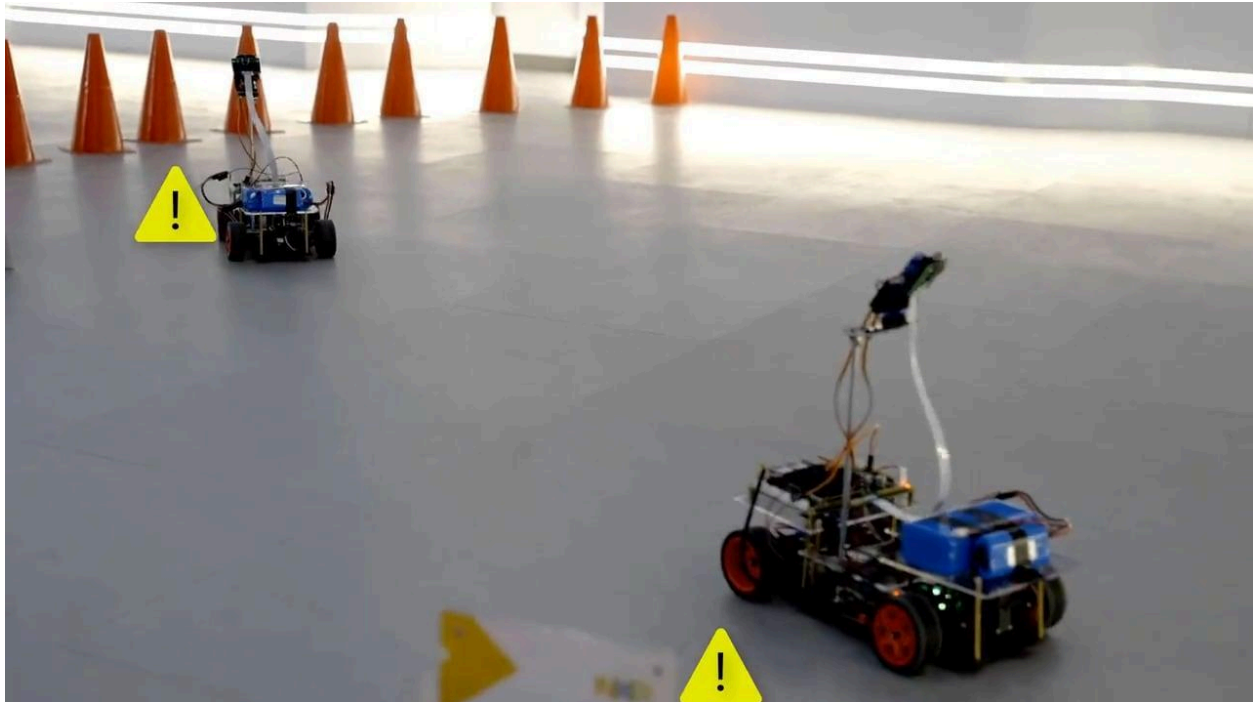
8. Autonomous Robot













Features

- Obstacle avoidance

- Path planning
- AI navigation
- Voice control

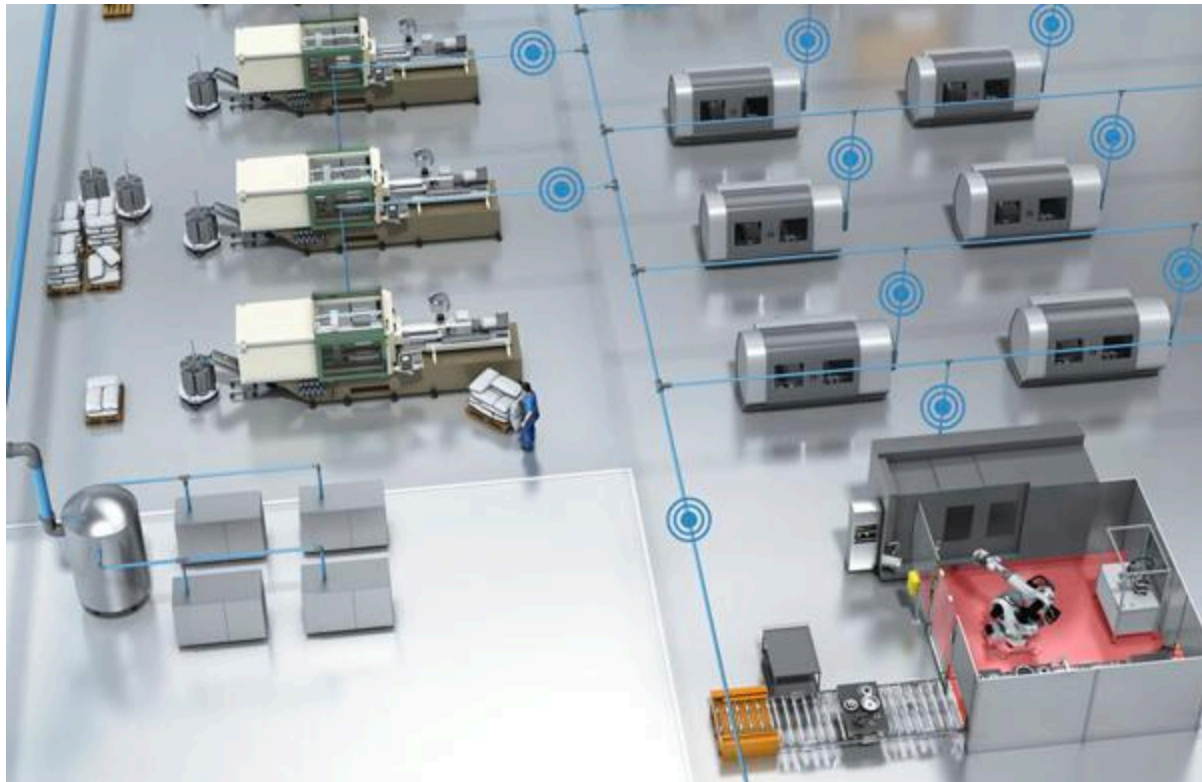
Technologies

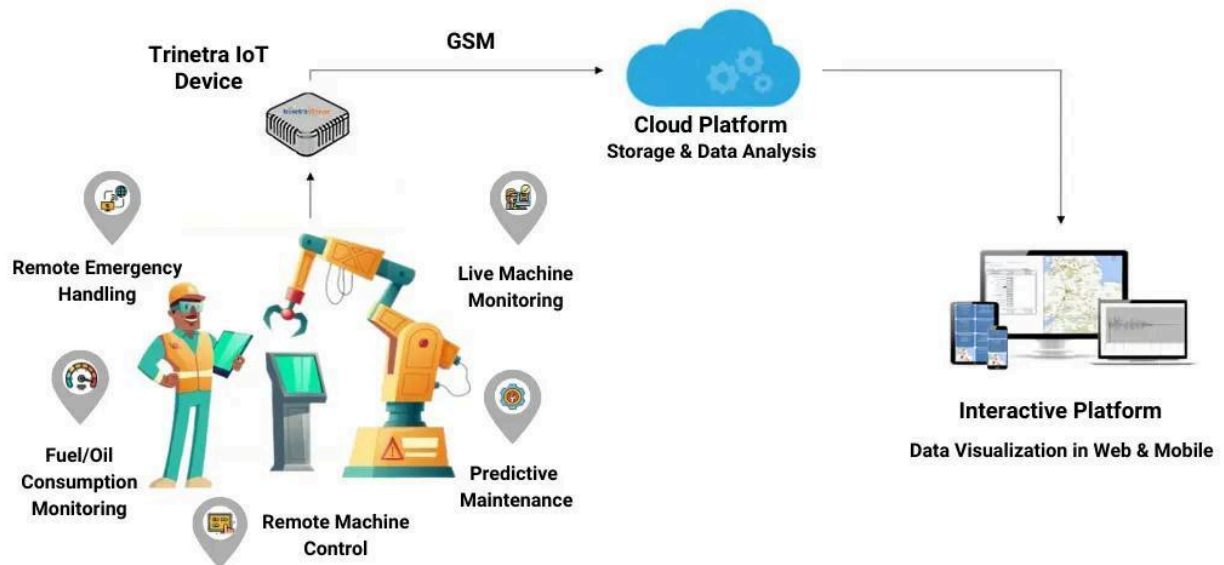
- ROS
- OpenCV
- AI/ML
- Sensors

9. Industrial Monitoring













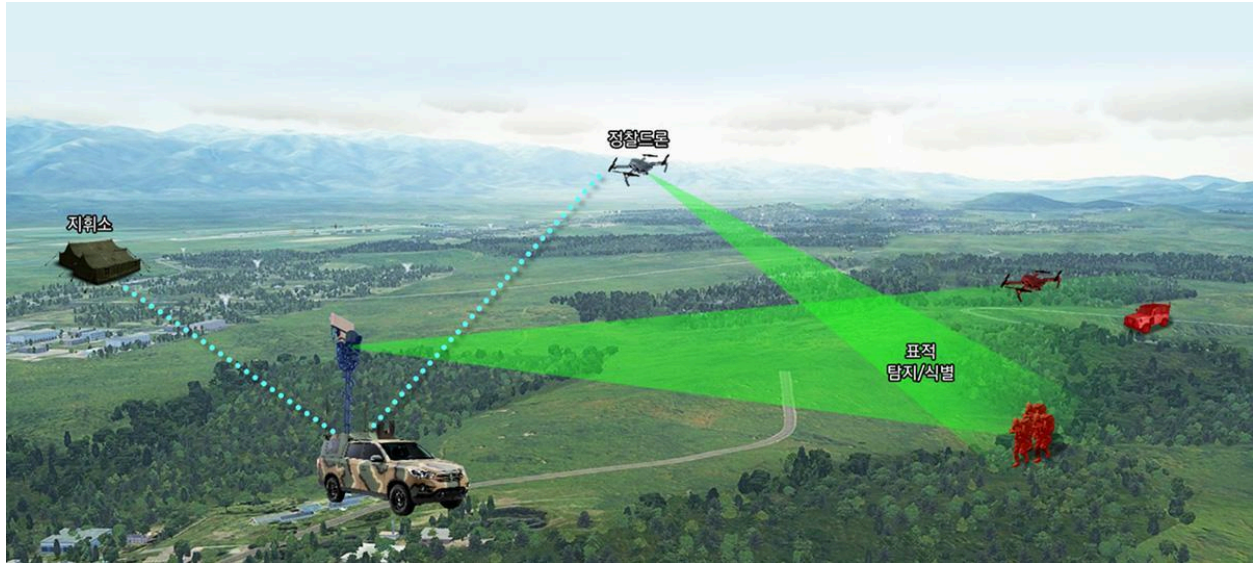
Features

- Machine temperature monitoring
- Predictive maintenance
- Gas leakage detection
- Real-time dashboard

Components

- ESP32
 - Gas Sensors
 - Vibration Sensors
 - Cloud Dashboard
-

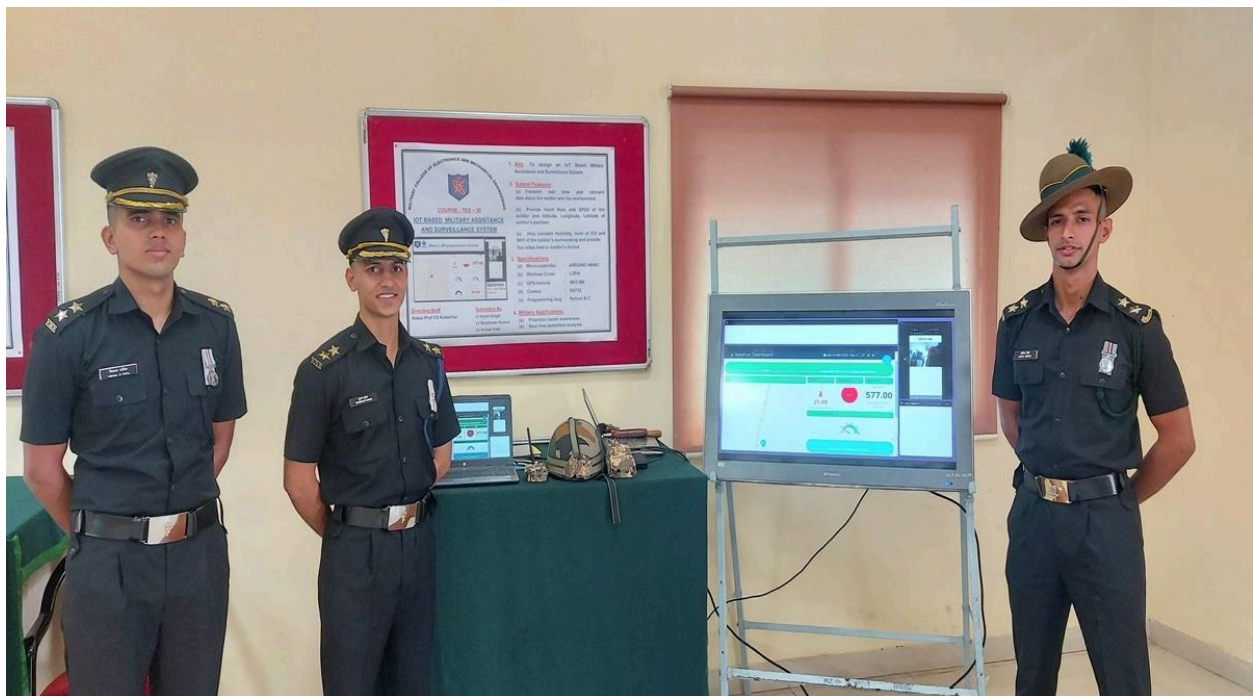
10. Defense Surveillance System













Features

- Night vision
- Thermal detection
- AI target identification

- Long-range communication

Advanced Technologies

- Thermal Imaging
 - Drone Integration
 - AI Detection
 - LoRa Mesh Network
-

Best Projects Based on Difficulty

Level	Projects
Beginner	Smart Helmet, LoRa Communication
Intermediate	Smart Agriculture, Industrial Monitoring
Advanced	AI CCTV, Autonomous Robot
Expert	Drone Swarm, Defense Surveillance

Most Powerful Career Combination

Embedded + AI + Wireless Networking + Robotics

This combination is highly useful in:

- Defense
- Aerospace
- Robotics
- Smart Cities
- Industrial Automation
- Autonomous Systems
- Surveillance Technology